

Constant-memory out-of-core single-cell preprocessing to one million cells with anndataoom

Zehua Zeng^{1,2} and anndata-oom contributors¹

¹OmicVerse Project

²starlitnightly@163.com | <https://github.com/omicverse/anndata-oom>

Abstract

Single-cell atlases now routinely exceed a million cells, but the standard analysis stack loads the entire expression matrix into memory, so peak RAM grows linearly with cell count and crashes commodity hardware. We present anndataoom, a Rust-backed out-of-memory AnnData store that keeps the matrix on disk and runs the omicverse preprocessing pipeline (quality control, normalization, scaling and principal component analysis) as lazy, chunked transforms, so peak memory is bounded by the chunk size rather than the number of cells. Across a Tabula Sapiens series spanning 5,000 to 1,053,033 cells under a 256 GB cap, anndataoom holds peak resident memory essentially flat at roughly 0.9 to 5.0 GB, whereas in-memory and scanpy-backed configurations grow memory linearly and are out-of-memory-killed beyond 228,032 cells. At the largest point where every configuration survives, anndataoom uses about 53x less memory than dense in-memory processing, and it is the only configuration that completes the million-cell dataset, finishing in 44.8 min at 5.0 GB peak. Wall-clock time tracks the in-memory band beyond the smallest dataset, and an optional GPU-mixed path is memory- and runtime-invariant. anndataoom delivers constant-memory, in-place, API-compatible out-of-core preprocessing at atlas scale on a workstation.

Introduction

Single-cell genomics has entered the atlas era, with reference resources such as Tabula Sapiens¹ and the CZ CELLxGENE Discover corpus^{2,3} aggregating tens to hundreds of millions of cells. The de facto data structure for these analyses is AnnData⁴, which organizes a central observations-by-variables expression matrix X together with aligned annotations, and the de facto algorithmic stack is built around SCANPY⁵ and integrative frameworks such as omicverse⁶. A canonical preprocessing pipeline—quality control, total-count normalization, log transformation, highly variable gene selection, scaling, and principal component analysis (PCA)—transforms raw counts into the low-dimensional embedding on which clustering, trajectory inference and visualization depend.

The central scalability obstacle is that this stack is fundamentally in-memory: X is materialized in RAM, so peak memory grows linearly with the number of cells. A dense million-cell matrix over the full gene set requires on the order of 10^2 GB, beyond the capacity of typical workstations and many shared compute nodes. As a result, the dataset size an analyst can process is dictated by available RAM rather than by the intrinsic cost of the computation.

Existing out-of-core strategies each relax this constraint at a cost. AnnData’s native backed mode reads X from disk without full materialization, but it can write back only X , loads obs/var eagerly, and frequently spills sparse data back into memory during processing. The experimental lazy-reading and Zarr-v3 facilities introduced in recent AnnData releases^{7,8} back arrays with Dask and xarray, but AnnData core implements no preprocessing algorithms; those live in SCANPY, where Dask support is partial and explicitly experimental—normalization and log1p are lazy, but highly variable gene selection and PCA force computation, and quality-control and scaling are not demonstrated as Dask-backed⁹. Consequently the full quality-control-to-PCA pipeline is not end-to-end out-of-core in the standard stack. Alternative ecosystems shift the burden elsewhere:

TileDB-SOMA and CELLxGENE Census^{3,10} require adopting a new store and streaming-query API rather than mutating `AnnData` in place; GPU frameworks such as `rapids-singlecell`¹¹ and `ScaleSC`¹² demand NVIDIA hardware and, at the largest scales, multi-GPU clusters; and the Rust fully-backed `anndata-rs` powering `SnapATAC2`^{13,14} is oriented toward single-cell epigenomics rather than general scRNA-seq preprocessing.

We introduce `anndataoom`¹⁵, an out-of-memory `AnnData` backend built on `anndata-rs`¹³ that fills the underserved niche of a true in-place, full-read/write, out-of-core engine that keeps the `SCANPY`/`omicverse` API and standard on-disk stores. `anndataoom` keeps `X` on disk and composes each preprocessing step as a lazy transform descriptor—per-cell size factors for normalization, a flag for `log1p`, a column index for gene selection, mean and standard-deviation vectors for scaling—that is evaluated on-the-fly during chunked reads. Normalization statistics are accumulated by a single-pass numerically stable update¹⁶, and PCA is computed by randomized SVD¹⁷ expressed as chunked matrix products, so the only in-memory products are the size- k working set and the final embedding. Peak RAM is therefore bounded by the chunk size, not the cell count.

We benchmark `anndataoom` against in-memory and `scanpy`-backed configurations of the `omicverse` pipeline across a `Tabula Sapiens` cell-count series from 5,000 to 1,053,033 cells under a 256 GB resident-memory cap. The headline result is constant-memory scaling: `anndataoom` holds peak memory roughly flat across the entire range while every in-memory alternative grows linearly and is out-of-memory-killed at atlas scale, yet `anndataoom` completes the full pipeline at near-identical wall-clock time, trading a small fixed startup cost for more than 50× memory savings and the ability to run datasets that crash every in-memory alternative.

Related work

The `AnnData` data model and its native out-of-core facilities. `AnnData`⁴ is the standard container for single-cell data, organizing a central matrix `X` alongside aligned annotations (`obs`, `var`, `obsm`, `varm`, `obsp`, `varp`, `layers`, `uns`) and persisting to chunked HDF5 (`.h5ad`) and Zarr stores. It offers a backed out-of-memory mode and, more recently, experimental lazy reading via `read_lazy/read_elem_lazy` and full Zarr-v3 support with sharding and pluggable cloud/GPU backends^{7,8,18}. Crucially, these facilities address I/O, lazy materialization, subsetting and on-disk concatenation only; `AnnData` core implements no preprocessing algorithms. Backed mode is also limited in practice: only `X` is written back, `obs/var` are read eagerly, and sparse reads can still spill to RAM. `anndataoom` instead provides a fully in-place backend in which preprocessing operations are evaluated against on-disk `X` through chunked, lazy transforms.

Distributed and lazy preprocessing in the `scanpy` stack. `SCANPY`⁵ supplies the preprocessing algorithms and offers experimental Dask support⁹: `normalize_total` and `log1p` are lazy, but highly variable gene selection and PCA trigger computation, and quality-control metrics and scaling are not demonstrated as Dask-backed. The documentation explicitly warns that Dask support is new, highly experimental, and safe only on documented functions, with manual chunk and worker tuning. The end-to-end quality-control-to-PCA pipeline is therefore not out-of-core in this stack. `anndataoom` targets exactly this pipeline as a single constant-memory path while preserving the `omicverse`⁶ `ov.pp` surface.

New-store and GPU approaches. `TileDB-SOMA`¹⁰ and `CZ CELLxGENE Discover Census`^{2,3} demonstrate out-of-core access to tens of millions of cells, including whole-corpus statistics on an 8 GB laptop, but require adopting a non-`AnnData` store and a streaming-query API; slices are converted to `AnnData` rather than mutated in place. GPU frameworks deliver large speedups: `rapids-singlecell`¹¹ is a near-drop-in `CuPy`-backed `SCANPY` replacement that now scales past a single GPU via multi-GPU Dask and managed-memory spilling, while `ScaleSC`¹² chunks along cells to reach 10–20M cells on a single A100 and circumvents the 32-bit indexing overflow that makes `rapids-singlecell` fail near 1.3M cells. Both require NVIDIA hardware. `anndataoom` is complementary: it is CPU-first and hardware-agnostic, and our benchmarks include an optional CPU–GPU-mixed path.

Rust-backed and curation layers. `anndata-rs`¹³ provides a fully-backed HDF5 `AnnData` that stays synchronized with disk, opens at near-zero memory, and processes matrices by chunk; it ships inside `SnapATAC2`¹⁴, which analyzes hundreds of thousands of cells in tens of GB but is oriented toward single-cell epigenomics. `anndataoom` builds on `anndata-rs` to deliver general scRNA-seq preprocessing. Orthogonally, data-management layers such as `LaminDB`¹⁹ and streaming data loaders such as `scDataset`²⁰ govern or stream large corpora for machine-learning training rather than performing in-place interactive preprocessing. All implementations rely on the scientific Python stack²¹.

Results

Benchmark design. We swept the omicverse `ov.pp` pipeline (load, quality control, preprocess, scale, PCA) across a seven-point Tabula Sapiens¹ cell-count series (Table 1), from TS-5k (5,000 cells) to TS-1M (1,053,033 cells, 60,606 genes), under a 256 GB resident-set-size (RSS) cap. Six configurations were compared: omicverse in-memory dense (`ov-anndata`), in-memory implicit-dense, scanpy backed mode (`backed='r'`), `anndataoom` (`ov-oom`), and the GPU-mixed variants of the out-of-memory and dense backends. Full per-dataset wall-clock and peak-RSS measurements are reported in Table 2.

Constant-memory scaling is the headline result. Peak memory is the decisive axis (Fig. 1b). The two `anndataoom` curves (CPU and GPU-mixed) coincide almost exactly and stay essentially flat at roughly 0.9–5.0 GB across the entire 5,000-to-1,053,033 range, while the in-memory and scanpy-backed configurations rise with unit slope, growing memory linearly with cell count. At TS-Epi (228,032 cells), the largest point where all four in-memory/backed configurations still survive, peak RSS was 104 GB for in-memory dense, 47 GB for the implicit-dense variant and 143 GB for scanpy-backed, versus only 2.0 GB for `anndataoom`—an approximately 53.0× memory reduction over dense in-memory processing. Beyond this point the in-memory configurations are progressively out-of-memory-killed: dense, scanpy-backed and the dense GPU-mixed variant fail at the first larger dataset, TS-Immune (592,317 cells; the implicit-dense variant survives there at roughly 592,317 cells but is killed at TS-1M), whereas only the two `anndataoom` curves complete the full series. At TS-1M (1,053,033 cells) `anndataoom` finishes in 44.8 min at a peak of 5.0 GB. The coincidence of the two `anndataoom` curves in both panels demonstrates mode-invariance: adding the GPU-mixed path changes neither memory footprint nor runtime.

Wall-clock time tracks the in-memory band beyond the smallest dataset. On end-to-end time (Fig. 1a), five of six configurations collapse onto a single near-linear log-log band; `anndataoom` pays a fixed startup overhead that makes it the slowest only at TS-5k, after which its curve converges with the in-memory band from TS-Vasc onward. scanpy-backed is the consistent time outlier, running roughly 2× faster at small and mid scale (for example, 375.8 s at TS-Epi versus ~800–924 s for the other surviving configurations), but it cannot reach the large datasets and is killed at TS-Immune. The per-stage decomposition (Fig. 2) shows preprocess as the dominant and fastest-growing stage, eventually swamping the others, while load and quality control shrink in relative share as data grows; `anndataoom`’s larger load/setup component at TS-5k accounts for its small-data overhead. At atlas scale this overhead is negligible: PCA alone on the million-cell dataset completed in 171 s (2.8 min). At TS-Immune and TS-1M, only the `anndataoom` bars are present; the in-memory and backed bars are replaced by OOM markers.

The CPU–GPU-mixed path is break-even. A direct comparison of CPU versus CPU–GPU-mixed execution (Fig. 3, Table 3) shows the two curves essentially superimposed for both backends, with mixed-mode speedups clustering tightly around unity (`anndataoom`: 0.96–1.08, aside from a 1.51 spike at TS-5k dominated by warm-up noise on a tiny dataset; in-memory dense: 0.90–1.01). GPU memory remained at 0 MB for all `anndataoom` mixed runs, indicating that PCA was not

Table 1. Benchmark datasets. Seven real Tabula Sapiens slices¹ from cellxgene² on the shared 60,606-gene panel, with layers["decontXcounts"] promoted to .X.

Dataset	Cells	Genes	On-disk (MB)	Source
TS-5k	5,000	60,606	130.1	Tabula Sapiens vasculature (sub)
TS-10k	10,000	60,606	222.5	Tabula Sapiens vasculature (sub)
TS-Vasc	42,650	60,606	2081.8	Tabula Sapiens vasculature
TS-Stromal	232,684	60,606	9594.1	Tabula Sapiens stromal compartment
TS-Epi	228,032	60,606	11066.5	Tabula Sapiens epithelial compartment
TS-Immune	592,317	60,606	18858.0	Tabula Sapiens immune compartment
TS-1M	1,053,033	60,606	15439.3	Tabula Sapiens stromal+epi+immune (concat)

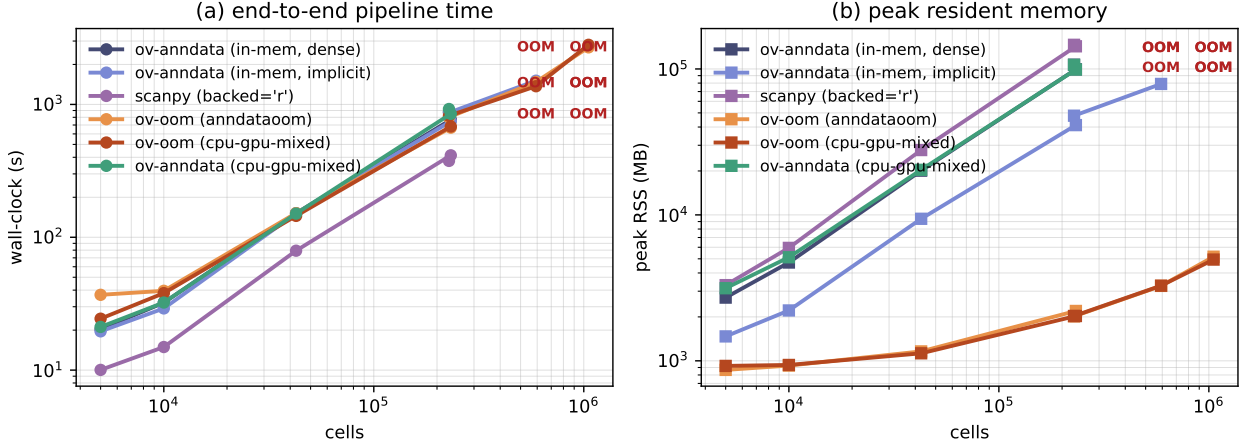


Figure 1. Fig. 1 | Constant-memory scaling. End-to-end pipeline wall-clock (left) and peak RSS (right) versus dataset size, log-log, for all six configurations. The two `ov-oom` curves (CPU and `cpu-gpu-mixed`) coincide and stay flat at ~1–5 GB, while the in-memory/backed configurations grow linearly and are killed at the 256 GB per-process cap for $\geq 592,317$ cells (“OOM”).

actually GPU-offloaded in these stages, whereas the dense mixed runs used 324–3,799 MB of GPU memory for no consistent time benefit. At TS-1M the mixed run took 2811 s versus 2689 s for CPU (speedup 0.96), at 4.8 GB versus 5.0 GB peak RSS—confirming that enabling the GPU path neither raises memory nor changes runtime for the out-of-memory backend.

Pipeline coverage. Of the 24 `ov.pp` functions surveyed, 14 are out-of-memory-adapted (Table 4), spanning the full quality-control-to-PCA pipeline plus neighbors, Leiden/Louvain, UMAP, t-SNE, MDE, normalization, `log1p` and robust-gene identification, with neighbors, UMAP and MDE GPU-capable. Functions that densify the matrix (highly variable feature selection, Pearson-residual normalization, regression) or perform backed variable lookups (Scrublet, cell-cycle scoring) remain unsupported, and `sude` fails in mixed mode (NaN). These limitations delineate the constant-memory operating envelope rather than detract from it: the core embedding pipeline that dominates atlas-scale preprocessing runs end-to-end out-of-core.

Discussion

Summary. `anndataoom` makes peak memory a function of chunk size rather than cell count, turning the dataset-size ceiling from a hardware constraint into a near-constant. Across a 5,000-to-1,053,033-cell Tabula Sapiens series under a 256 GB cap, it held peak RSS essentially flat at roughly 0.9–5.0 GB, used about $53.0\times$ less memory than dense in-memory processing at the largest commonly survivable dataset, and was the only configuration to complete the million-cell pipeline (44.8 min, 5.0 GB peak) while every in-memory and backed alternative was out-of-memory-killed beyond 228,032 cells. Critically, this memory saving came at near-identical wall-clock time, trading only a

Table 2. End-to-end wall-clock (s) and peak RSS (MB) for each (configuration, dataset) cell. “*” = implicit-centered scaling; “G” = cpu-gpu-mixed. “OOM” marks runs killed by the 256 GB cap.

	ov-anndata		ov-anndata*		scanpy-backed		ov-oom		ov-oom ^G		ov-anndata ^G	
	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)	t (s)	RSS (MB)
TS-5k	20.3	2711.5	19.6	1470.1	10.0	3300.6	36.9	866.7	24.4	922.9	21.1	3144.1
TS-10k	32.2	4716.8	29.1	2213.7	14.9	5940.4	39.5	927.8	38.0	936.9	32.3	5124.2
TS-Vasc	151.7	20095.3	151.2	9404.1	79.2	27821.5	151.9	1160.2	144.9	1125.8	150.7	20365.2
TS-Stromal	766.9	99371.0	745.9	41229.6	413.9	142962.9	668.5	2196.2	682.3	2028.0	849.1	99104.4
TS-Epi	880.1	106938.9	866.9	47831.9	375.8	146379.4	800.7	2017.6	821.5	2028.7	923.7	106605.5
TS-Immune	OOM	–	1509.0	78949.4	OOM	–	–	1476.6	3261.8	1371.0	3272.7	OOM
TS-1M	OOM	–	OOM	–	OOM	–	2688.7	5168.7	2810.7	4940.9	OOM	–

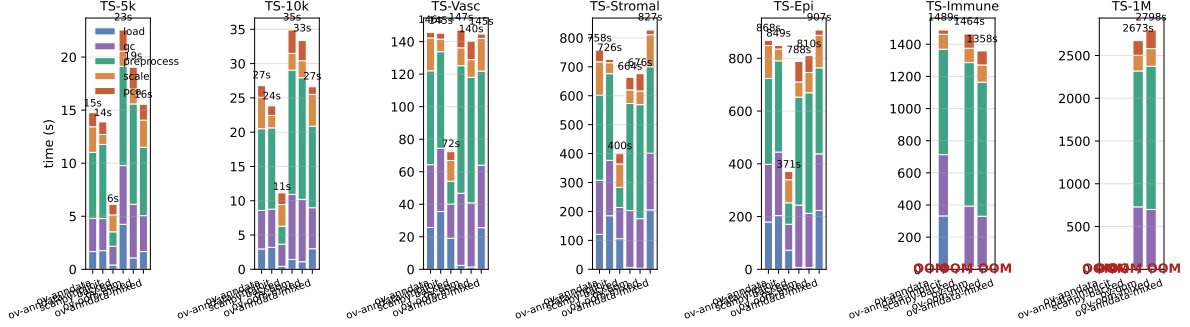


Figure 2. Fig. 2 | Per-stage wall-clock breakdown across all benchmark points. **preprocess** dominates and grows fastest; **ov-oom’s normalize** is effectively free, amortised into the next chunked read. Beyond TS-Epi only the **ov-oom** bars remain.

small fixed startup cost. Because preprocessing steps are encoded as lazy transform descriptors over on-disk X and PCA is computed by chunked randomized SVD¹⁷ with single-pass statistics¹⁶, the backend remains a drop-in within the omicverse⁶ and AnnData⁴ ecosystem rather than a separate store.

Positioning. anndataoom occupies a niche left open by existing approaches: a true in-place, full-read/write out-of-core backend that keeps the SCANPY/omicverse API and standard on-disk stores, without forcing a new store and query API (TileDB-SOMA/Census^{3,10}), GPUs (rapids-singlecell, ScaleSC^{11,12}), the experimental and partial Dask path⁹, or an epigenomics-oriented Rust stack¹⁴. It is complementary to all of these: the constant-memory CPU engine can serve as the substrate that feeds GPU acceleration or curation layers¹⁹ when that hardware or governance is available.

Limitations and outlook. The constant-memory envelope is bounded to functions that avoid densifying X or eager backed variable lookups; densifying operations such as Pearson-residual normalization and regression, and lookup-heavy steps such as Scrublet and cell-cycle scoring, remain in-memory, and **sude** is unstable in mixed mode. In the configurations tested, the GPU-mixed path was break-even and did not actually offload PCA (GPU memory stayed at 0 MB for anndataoom), so realizing GPU speedups for the chunked SVD is the clearest avenue for future work, alongside extending the out-of-memory adaptation to the remaining densifying steps and exploiting Zarr-v3 sharded and remote stores⁸ for cloud-native execution. More broadly, by decoupling processable dataset size from RAM, anndataoom lets a commodity workstation run preprocessing pipelines that previously required large-memory servers, lowering the hardware barrier to atlas-scale single-cell analysis.

Methods

Storage and transform layer. The h5ad reader is the **anndata-rs** Rust library¹³, exposed to Python as a **BackedArray** proxy that yields **scipy.sparse** chunks without densifying on read.

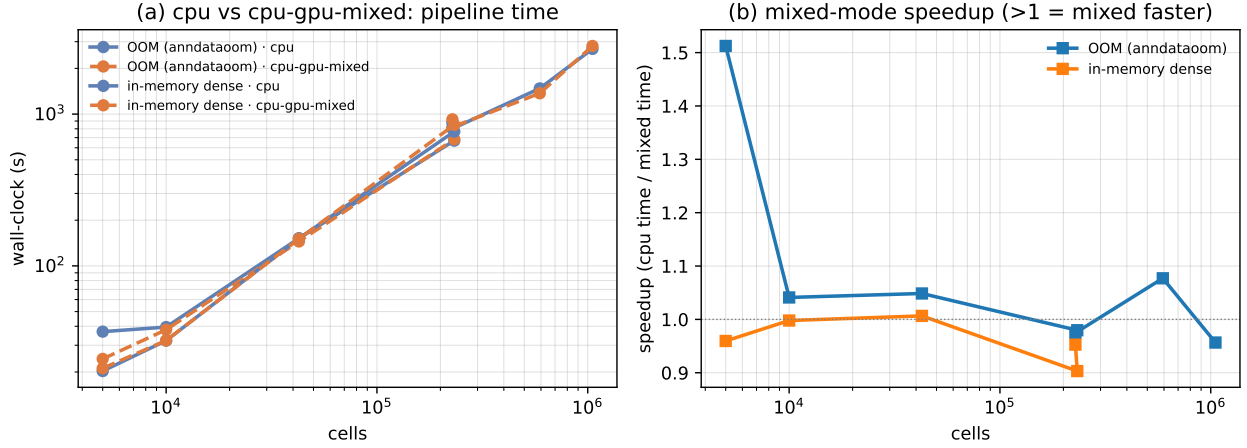


Figure 3. Fig. 3 | CPU vs CPU-GPU-mixed. Pipeline wall-clock (left) and speedup ratio $t_{\text{cpu}}/t_{\text{mix}}$ (right). Both backends’ curves are near-superimposed; mixed-mode speedups cluster around 1.0, confirming mode-invariance.

Table 3. CPU vs CPU-GPU-mixed on the out-of-memory and in-memory backends. “GPU (MB)” is peak PyTorch device memory; ≈ 0 for the out-of-memory backend (its PCA path is CPU-resident).

Backend	Dataset	cpu		cpu-gpu-mixed			speedup $t_{\text{cpu}}/t_{\text{mix}}$
		t (s)	RSS (MB)	t (s)	RSS (MB)	GPU (MB)	
OOM (anndataoom)	TS-5k	36.9	866.7	24.4	922.9	0.0	1.51
OOM (anndataoom)	TS-10k	39.5	927.8	38.0	936.9	0.0	1.04
OOM (anndataoom)	TS-Vasc	151.9	1160.2	144.9	1125.8	0.0	1.05
OOM (anndataoom)	TS-Stromal	668.5	2196.2	682.3	2028.0	0.0	0.98
OOM (anndataoom)	TS-Epi	800.7	2017.6	821.5	2028.7	0.0	0.97
OOM (anndataoom)	TS-Immune	1476.6	3261.8	1371.0	3272.7	0.0	1.08
OOM (anndataoom)	TS-1M	2688.7	5168.7	2810.7	4940.9	0.0	0.96
in-memory dense	TS-5k	20.3	2711.5	21.1	3144.1	324.4	0.96
in-memory dense	TS-10k	32.2	4716.8	32.3	5124.2	400.7	1.00
in-memory dense	TS-Vasc	151.7	20095.3	150.7	20365.2	898.9	1.01
in-memory dense	TS-Stromal	766.9	99371.0	849.1	99104.4	3798.6	0.90
in-memory dense	TS-Epi	880.1	106938.9	923.7	106605.5	3728.1	0.95

Per-cell normalisation is left multiplication by a sparse diagonal of inverse size factors; $\log(1+x)$ is applied to the stored values only (sparsity-preserving, since $\log(1+0) = 0$). Per-gene mean and variance are computed in a single chunked pass with a hybrid estimator: per-chunk statistics use the sparse-native $E[X^2] - E[X]^2$ identity and cross-chunk accumulation uses Welford’s method¹⁶, keeping the error at the `float32`→`float64` floor ($\sim 7 \times 10^{-8}$ relative).

Chunked PCA. Dimensionality reduction uses randomised SVD¹⁷ over the lazy scaled matrix. When the post-HVG matrix fits a configurable budget it is built once into RAM and factorised in a single pass; otherwise an implicit-centered formulation rewrites the matrix–vector products through $(N - \mu)/\sigma \cdot W = N \cdot \text{diag}(1/\sigma) \cdot W - (\mu/\sigma) \cdot W$, where N is the sparse normalise+ $\log(1+x)$ view, so no dense per-chunk block is allocated. When materialising, HVG columns are sliced from the sparse parent *before* densifying—normalisation is per-row and $\log(1+x)$ element-wise, so the column slice commutes through both—so only a $(\text{chunk} \times n_{\text{HVG}})$ block is ever dense, numerically identical to densifying the full panel (maximum absolute difference $\sim 10^{-7}$).

Benchmark configurations. Four CPU storage/scaling backends: `ov-anndata` (`anndata.read_h5ad`; `ov.pp.scale` densifies the full panel); `ov-anndata-implicit` (implicit-centered scaled wrapper); `scanpy-backed` (`backed='r'`, then the standard scanpy pipeline—the first quality-control op has no backed handler, so the workflow materialises to memory before QC, leaving the backed advantage

Table 4. omicverse.pp compatibility on the out-of-memory backend. ✓ runs to completion; ✗ raises; superscript ^G marks GPU offload under `cpu-gpu-mixed`.

ov.pp function	cpu	mixed	GPU-offload	note
qc	✓	✓	—	
preprocess	✓	✓	—	
scale	✓	✓	—	
pca	✓	✓	—	
neighbors	✓	✓ ^G	yes	
leiden	✓	✓	—	
louvain	✓	✓	—	
umap	✓	✓ ^G	yes	
tsne	✓	✓	—	
mde	✓ ^G	✓ ^G	yes	torch (GPU-capable)
sude	✓	✗	—	fails in mixed (NaN)
normalize_total	✓	✓	—	
log1p	✓	✓	—	
identify_robust_genes	✓	✓	—	
highly_variable_features	✗	✗	—	pegasus HVF densifies
highly_variable_genes	✗	✗	—	standalone; use <code>preprocess</code>
normalize_pearson_residuals	✗	✗	—	not OOM-adapted
regress	✗	✗	—	scanpy <code>regress_out</code> densifies
scrublet	✗	✗	—	backed var lookup
score_genes_cell_cycle	✗	✗	—	backed var lookup
filter_cells	✗	✗	—	not OOM-adapted
filter_genes	✗	✗	—	not OOM-adapted
anndata_to_GPU	✗	✗	—	needs rapids
anndata_to_CPU	✗	✗	—	needs rapids

only at load); and `ov-oom` (`anndataoom.read`; operations dispatch to the chunked path via `is_oom`). Two `cpu-gpu-mixed` variants (`ov-oom-mixed`, `ov-anndata-mixed`) run the identical pipeline with PCA/neighbours/UMAP offloaded to a GPU via PyTorch.

Datasets and pipeline. Seven Tabula Sapiens slices¹ from cellxgene² on the 60,606-gene panel: TS-Vasculature (42,650) and two seed-0 subsamples (5,000/10,000); the stromal (232,684), epithelial (228,032) and immune (592,317) compartments; and a one-million-cell concatenation (1,053,033). Each uses `layers["decontXcounts"]` as `.X`; the harness promotes the raw-counts layer when `.X` appears log-normalised. The pipeline is `ov.pp.qc` (mode `seurat`, doublets off), `ov.pp.preprocess` (`shiftlogpearson`, 2,000 HVGs), `ov.pp.scale` (`max_value=10`) and `ov.pp.pca` (50 components), all on one consistent software stack.

Measurement. Per-stage wall-clock and RSS are captured around each stage (peak RSS = maximum post-stage RSS). The main sweep runs on a single CPU node under a 256 GB per-process cap; the GPU-mixed comparison runs on an NVIDIA H100, additionally recording per-stage peak PyTorch device memory. Numerical equivalence between backends is quantified by per-component absolute cosine similarity.

Code and data availability. The harness, figure/table generators and dataset cards are at the project repository¹⁵. Subsamples use a fixed seed; the million-cell input is the row-concatenation of the three compartment slices.

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